Library SIP standard

xx/checkIn

xx/checkOut

Checkout

This message is used by the SC to request to check out an item, and also to cancel a Checkin request that did not successfully complete. The ACS must respond to this command with a Checkout Response message.

11<SC renewal policy><no block><transaction date><nb due date><institution id><patron identifier><item identifier><terminal password><patron password><item properties><fee acknowledged><cancel>

	Field	ID	Format
	SC renewal policy		1-char, fixed-length required field: Y or N.
	no block		1-char, fixed-length required field: Y or N.
	transaction date		18-char, fixed-length required field: YYYYMMDDZZZZHHMMSS. The date and time that the patron checked out the item at the SC unit.
	nb due date		18-char, fixed-length required field: YYYYMMDDZZZZHHMMSS
	institution id	AO	variable-length required field
	patron identifier	AA	variable-length required field
	item identifier	AB	variable-length required field
	terminal password	AC	variable-length required field
2.00	*item properties*	*CH*	*variable-length optional field*
2.00	*patron password*	*AD*	*variable-length optional field*
2.00	*fee acknowledged*	*BO*	*1-char, optional field: Y or N*
2.00	*cancel*	*BI*	*1-char, optional field: Y or N*

Checkin

This message is used by the SC to request to check in an item, and also to cancel a Checkout request that did not successfully complete. The ACS must respond to this command with a Checkin Response message.

09<no block><transaction date><return date><current location><institution id><item identifier><terminal password><item properties><cancel>

	Field	ID	Format
	no block		1-char, fixed-length required field: Y or N.
	transaction date		18-char, fixed-length required field: YYYYMMDDZZZZHHMMSS
	return date		18-char, fixed-length required field: YYYYMMDDZZZZHHMMSS
	current location	AP	variable-length required field
	institution id	AO	variable-length required field
	item identifier	AB	variable-length required field
	terminal password	AC	variable-length required field
2.00	*item properties*	*CH*	*variable-length optional field*
2.00	*cancel*	*BI*	*1-char, optional field: Y or N*

xx/login

2.00 Login

This message can be used to login to an ACS server program. The ACS should respond with the Login Response message. Whether to use this message or to use some other mechanism to login to the ACS is configurable on the SC. When this message is used, it will be the first message sent to the ACS.

93<UID algorithm><PWD algorithm><login user id><login password><location code>

	Field	ID	Format
2.00	*UID algorithm*		*1-char, fixed-length required field; the algorithm used to encrypt the user id.*
2.00	*PWD algorithm*		*1-char, fixed-length required field; the algorithm used to encrypt the password.*
2.00	*login user id*	*CN*	*variable-length required field*
2.00	*login password*	*CO*	*variable-length required field*
2.00	*location code*	*CP*	*variable-length optional field; the SC location.*

xx/item (Book information)

2.00 Item Information

This message may be used to request item information. The ACS should respond with the Item Information Response message.

17<transaction date><institution id>< item identifier ><terminal password>

Field	ID	Format
transaction date		18-char, fixed-length required field: YYYYMMDDZZZZHHMMSS
institution id	AO	variable-length required field
item identifier	AB	variable-length required field.
terminal password	AC	variable-length optional field

Xx/logout

2.00 End Patron Session

This message will be sent when a patron has completed all of their transactions. The ACS may, upon receipt of this command, close any open files or deallocate data structures pertaining to that patron. The ACS should respond with an End Session Response message.

35<transaction date><institution id><patron identifier><terminal password><patron password>

Field	ID	Format
transaction date		18-char, fixed-length required field: YYYYMMDDZZZZHHMMSS
institution id	AO	variable-length required field
patron identifier	AA	variable-length required field.
terminal password	AC	variable-length optional field
patron password	AD	variable-length optional field

Aside from xx/item which return the books array matching the book info. Other API should all return Json as below.

{ status:1} // if ok

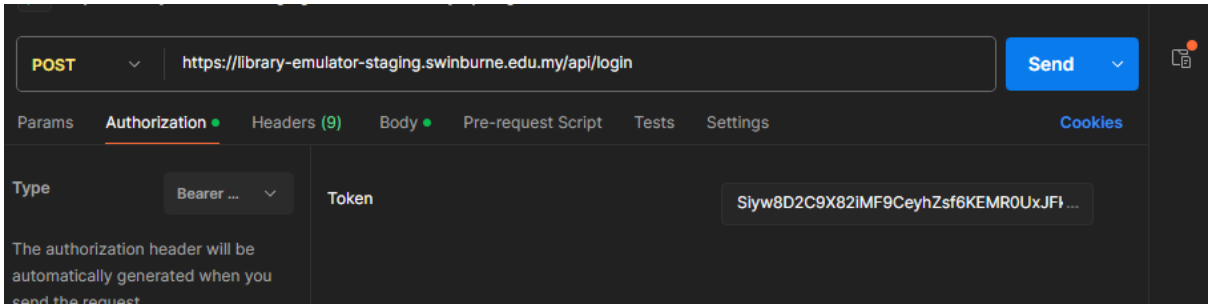
{ status:0} // if nothing happens. Unlikely in most API.

{ status:-1} // if error. Unlikely since no checking.

Working with API and samples

<u>API Key</u>

Siyw8D2C9X82iMF9CeyhZsf6KEMR0UxJFkZB8xjWw5LIpLOaLtnA5GZ5GvMsSAEEOu085IWuH4miW036dMLEgWwFrlB2THd2zfr4h4UmbkueYshuuIFwxbbRWc1d1Ssq

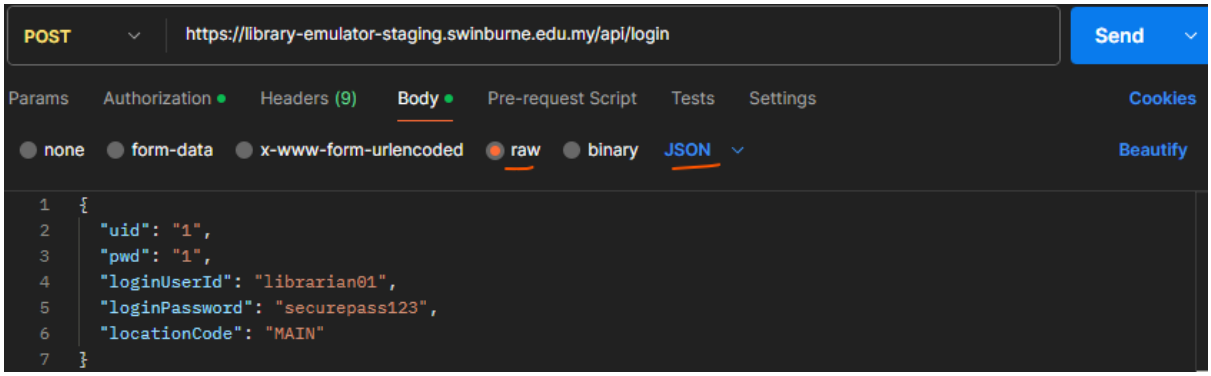


To interact with the API, send an **HTTP POST** request to:

https://library-emulator-staging.swinburne.edu.my/api/xxAPI

Replace xxAPI with the desired endpoint (e.g., login, checkout, etc.).

- **Authorization**: Use **Bearer Token** authentication with your API key.
- **Body**: Always send the payload as **raw JSON** in the body of the request.



checkOut

{

"scRenewalPolicy": "Y",

"noBlock": "N",

"transactionDate": "202501180800000000",

"nbDueDate": "202501250800000000",

"institutionId": "LIB001",

"patronIdentifier": "P123456",

"itemIdentifier": "I789012",
"terminalPassword": "term123",
"itemProperties": "Hardcover, Fiction",
"patronPassword": "patron456",
"feeAcknowledged": "Y",
"cancel": "N"
}

checkIn

{
"noBlock": "N",
"transactionDate": "202501180800000000",
"returnDate": "202501180800000000",
"currentLocation": "Main Library",
"institutionId": "LIB001",
"itemIdentifier": "I789012",
"terminalPassword": "term123",
"itemProperties": "Hardcover, Fiction",
"cancel": "N"
}

login

{
"uid": "1",
"pwd": "1",
"loginUserId": "librarian01",
"loginPassword": "securepass123",
"locationCode": "MAIN"

}

item

{
  "transactionDate": "202501180800000000",
  "institutionId": "LIB001",
  "itemIdentifier": "I789012",
  "terminalPassword": "term123"
}

logout

{
  "transactionDate": "202501180800000000",
  "institutionId": "LIB001",
  "patronIdentifier": "P123456",
  "terminalPassword": "term123",
  "patronPassword": "patron456"
}

Check your log:

Link
https://library-emulator-staging.swinburne.edu.my/